My Thesis Title

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Artificial Intelligence and
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Bachelor's Thesis

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Abstract

Accurately quantifying uncertainty in Large Language Models (LLMs) is crucial for their deployment in safety-critical domains. However, standard uncertainty estimation methods typically rely on a single probability distribution defined by fixed decoding hyperparameters. We argue that this "point-estimate" approach fails to adequately capture epistemic uncertainty, often conflating benign linguistic diversity with factual ignorance. In this thesis, we introduce *Credal Semantic Entropy*, a rigorous framework for hallucination detection grounded in the theory of Imprecise Probabilities. We propose a method to construct Credal Sets—convex sets of probability distributions—based on the statistical notion of relative likelihood. Instead of relying on a single temperature, we identify a set of statistically valid temperatures that maximize the likelihood of calibration data. We then aggregate generated sequences into semantic clusters and derive interval-valued probabilities for each meaning. By computing the Upper Semantic Entropy over these intervals, we obtain a worst-case estimate of uncertainty that is robust to hyperparameter volatility.

RESULTS ARE MISSING

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1 Introduction

1.1 Motivation

Large Language Models (LLMs) have fundamentally transformed the landscape of Artificial Intelligence, demonstrating unprecedented capabilities in natural language understanding, reasoning and generation. LLMs are increasingly deployed in critical domains ranging from healthcare [Bed+25] to legal analysis [Sii+25]. However, despite their fluency, these models suffer from a critical reliability failure: the tendency to generate plausible-sounding but factual incorrect statements, a phenomenon widely known as hallucinations [Hua+23].

For high-stakes applications, accuracy alone is insufficient. A system must also be self-aware of its limitations [Bom+22]. A doctor or lawyer using an LLM needs to know when the model is uncertain so they can verify the output. Unfortunately, modern LLMs are frequently "confidently wrong", assigning high probability scores to their hallucinations due to poor calibration [Guo+17; DD20] and the nature of their training objectives [Ji+23].

Standard uncertainty metrics, such as predictive entropy [MG21] or lexical similarity [Fom+20], often fail to detect these errors because they rely on a single, precise probability distribution. This "point-estimate" approach mathematically conflates two distinct sources of uncertainty [HW21]: *aleatoric uncertainty* (the natural diversity of valid human language) and *epistemic uncertainty* (the model's lack of knowledge). When an LLM distributes its probability mass across five different ways of saying the correct answer, standard metrics flag this as high uncertainty, confusing benign linguistic flexibility with factual ignorance.

1.2 Problem Statement

The core problem addressed in this thesis is the unreliability of single-point probability estimates for detecting hallucinations in free-form Question Answering.

Current state-of-the-art methods, such as Semantic entropy [KGF23], attempt to mitigate this by grouping lexically diverse answers into semantic clusters. However, even Semantic Entropy relies on a single, static probability distribution generated at a fixed temperature. If this underlying distribution is miscalculated, i.e., peaked

around an incorrect fact, the metric will fail to signal a warning. To robustly detect hallucinations, we need a framework that moves beyond precise probabilities and considers a set of plausible model states.

1.3 Contributions

This thesis proposes a novel methodology for quantifying epistemic uncertainty by bridging the gap between Imprecise Probabilities and Large Language Model decoding.

Drawing on the theory of *Credal Prediction based on Relative Likelihood* [Löh+25], we introduce a statistically rigorous method to construct a Credal Set of semantic distributions. Our specific contributions are as follows:

- **Valid Temperature Selection via Relative Likelihood**: We propose a method to identify a statistically valid range of decoding temperatures. By treating temperatures as a hyperparameter and applying an α-cut, we systematically discard implausible distributions.

- **Construction of Semantic Credal Sets**: We extend the concept of Semantic Entropy to operate over Interval-Valued Probabilities. By computing the minimum and maximum probability of semantic clusters across valid temperatures, we construct a convex credal set that captures the model's true epistemic ambiguity.

- **Empirical Validation**:

1.4 Thesis Structure

The remainder of this thesis is organized as follows:

- **Chapter 2 Background**: establishes the theoretical foundations necessary for this work. It reviews the architecture of Large Language Models, explains autoregressive decoding strategies, and introduces the mathematical frameworks of Imprecise Probabilities and Credal Sets.

- **Chapter 3 Related Work**: surveys the current landscape of uncertainty quantification. It contrasts traditional point-estimate methods with recent advancements in semantic consistency, specifically focusing on Semantic Entropy as the primary baseline for our research.

- **Chapter 4 methodology**: formalizes our novel contribution: *Credal Semantic Entropy*. It details the algorithm for selecting valid temperatures via relative likelihood and deriving robust uncertainty bounds from interval-valued probability distributions.

- **Chapter 5 Experiments**: presents the experimental setup and results, analyzing the performance of our method against strong baselines.

- **Chapter 6 Conclusion**:

2 Background

This chapter establishes the theoretical and methodological foundations necessary to understand the contributions of this thesis. Our work lies at the intersetion of Natural Language Processing (NLP) and robust Uncertainty Quantification (UQ). To rigorously address the problem of hallucination detection, we must first understand the probabilistic nature of modern language models and the limitations of traditional statistical tools.

2.1 Large Language Models and Generation

Large Language Models (LLMs) represent a class of deep learning models designed to process, understand, and generate human language. At their core, modern LLMs are built upon the Transformer architecture, utilizing mechanisms such as self-attention to model complex dependencies between words across long sequences [Vas+17; Nav+24]. While they demonstrate capabilities in reasoning [Wei+23] and coding [Tia+23], their fundamental operation remains probabilistic. They predict the likelihood of the next token in a sequence based on the preceding context [Vas+17; BCB16].

2.1.1 Transformers and Next-token prediction

The primary objective of an autoregressive LLM is to model the probability distribution of a target sequence of tokens $\mathbf{y} = (y_1, y_2, \ldots, y_T)$ given an input context or prompt x. The model decomposes the joint probability of the sequence into a product of conditional probabilities using the chain rule of probability [BCB16]:

$$P(\mathbf{y}|x) = \prod_{t=1}^{T} P(y_t \mid \mathbf{y}_{<t}, x) \tag{2.1}$$

Where y_t is the token at step t and $\mathbf{y}_{<t}$ represents all preceeding tokens $y_1, \ldots, y_{t-1}$ in the generated sequence. To compute these probabilities, the Transformer processes the input x and the generated history $\mathbf{y}_{<t}$ to produce a vector of non-normalized scores, known as logits, for every token in the vocabulary $\mathcal{V}$. Let z_i be the logit

corresponding to the i-th token $v_i \in \mathcal{V}$. The model converts these logits into a valid probability distribution using the Softmax function [Bri90; Vas+17]:

$$P(y_t = v_i \mid \mathbf{y}_{<t}, x) = \frac{\exp(z_i)}{\sum_{j=1}^{|\mathcal{V}|} \exp(z_j)} \qquad \forall i \in 1, \ldots, |\mathcal{V}| \tag{2.2}$$

This process is repeated iteratively, where the token selected at step t is appended to the input for step $t+1$ [Vas+17].

2.1.2 Decoding Strategies and the Role of Temperature

Once the probability distribution over the Vocabulary is computed for the next token, a **decoding strategy** determines what token will be selected. Common maximization-based strategies include **Greedy Decoding** or **Beam Search**. **Greedy Decoding** is the simplest deterministic method. It selects the token with the highest probability mass at the current time step t [SVL14]:

$$y_t = \arg\max_{y \in \mathcal{V}} P(y \mid \mathbf{y}_{<t}, x) \tag{2.3}$$

While computationally efficient, this strategy often yields sub-optimal results. By focusing solely on local probability maximization, the strategy risks settling into local optima, effectively missing high-probability sequences hidden behind lower-probability initial tokens. Furthermore, greedy decoding is known to produce repetitive and generic text, as it fails to account for the diversity and global coherence of the sequence [WR16; Hol+20].

Beam Search extends Greedy Decoding by maintaining multiple hypotheses at each time step, rather than a single sequence. It introduces a hyperparameter k, known as the **beam width**, which determines the number of candidates to keep. At each step t, the algorithm expands all k candidates from the previous step by appending every possible next token from the vocabulary $\mathcal{V}$ [FA17].

Formally, let $\mathcal{B}_{t-1} = \{\mathbf{y}_{<t}^{(1)}, \ldots, \mathbf{y}_{<t}^{(k)}\}$ be the set of k hypothesis sequences generated up to step $t-1$. The algorithm proceeds in three steps [FA17]:

1. **Expansion**: Each hypothesis in $\mathcal{B}_{t-1}$ is extended with every possible token v from the vocabulary $\mathcal{V}$, creating a candidate set $\mathcal{C}_t$.

2. **Scoring**: Each candidate sequence is assigned a score based on its cumulative log-probability. The score for a sequence $\mathbf{y}_{1:t}$ (representing $y_1, \ldots, y_t$) is defined recursively:

$$\text{Score}(\mathbf{y}_{1:t}) = \text{Score}(\mathbf{y}_{1:t-1}) + \log P(y_t \mid \mathbf{y}_{<t}, x) \tag{2.4}$$

3. **Selection (Pruning)**: The algorithm selects the k candidates with the highest scores to form the new hypothesis set $\mathcal{B}_t$:

$$\mathcal{B}_t = \underset{\mathbf{y}_{1:t} \in \mathcal{C}_t}{\text{arg-top-}k} \left(\text{Score}(\mathbf{y}_{1:t})\right) \tag{2.5}$$

This approach allows the model to recover from locally sub-optimal decisions if they lead to a highly probable global sequence. However, Beam Search is computationally more expensive than greedy methods and, as noted by Holtzman et al. [Hol+20], can still suffer from repetitive loops or generic responses in open-ended generation tasks.

Sampling represents a fundamental shift from maximization-based decoding strategies towards stochastic generation. Instead of strictly selecting the token with the highest probability, the model randomly draws the next token y_t from the conditional probability distribution $P(y_t \mid \mathbf{y}_{<t}, x)$.

Formally, at each time step t, the next token is sampled according to:

$$y_t \sim P(y \mid \mathbf{y}_{<t}, x) \tag{2.6}$$

This stochasticity allows the model to generate more diverse and creative outputs for the same input. However, pure random sampling from the full distribution can occasionally lead to incoherence. If the "tail" of the distribution, that contains very low-probability tokens, is sampled, it may disrupt the semantic flow or factual accuracy of the sequence. To mitigate this risk while preserving diversity, two truncation techniques are commonly employed:

1. **Top-k Sampling** [FLD18]: The model samples from the top k most probable tokens. This effectively truncates the unreliable tail of the distribution.

2. **Nucleus (Top-p) Sampling** [Hol+20]: The model samples from the smallest set of tokens whose cumulative probability exceeds a threshold p (e.g., $p = 0.9$). Unlike Top-k, which uses a fixed number of tokens, Top-p adapts dynamically. The set of candidates grows when the model is uncertain (flat distribution) and shrinks when the model is confident (peaked distribution).

While these truncation methods improve coherence, they operate by removing options. A more continuous method for controlling the shape of the probability distribution, and the core mechanism for this thesis, is **Temperature Scaling**.

Temperature Scaling is a technique used to calibrate the confidence of the model's predictions. Unlike Top-k or Nucleus Sampling, which truncate the vocabulary, temperature scaling modifies the probability distribution itself by rescaling the logits before the Softmax function is applied [HVD+15].

Let z_i be the logit for token v_i. We introduce a hyperparameter $\tau > 0$, called **temperature**. The probability of selecting token v_i is then given by:

$$P(y_t = v_i \mid \mathbf{y}_{<t}, x; \tau) = \frac{\exp(z_i/\tau)}{\sum_{j=1}^{|\mathcal{V}|} \exp(z_j/\tau)} \tag{2.7}$$

The value of τ controls the entropy of the ouptut distribution. When choosing a **low temperature**, $\tau < 1.0$, the distribution becomes sharper, i.e., more peaked. The difference between large and small logits is exaggerated, causing the model to become more confident and deterministic. In the limit, this approximates greedy decoding. Using a **high temperature**, $\tau > 1.0$, flattens the distribution towards a uniform distribution. This increases the likelihood of selecting lower-probability tokens, resulting in more diverse but also potentially more incoherent or hallucinatory outputs. For the temperature $\tau = 1.0$, the distribution remains unchanged, reflecting the model's original logits [HVD+15].

In standard generation tasks, τ is typically a fixed hyperparameter tuned to balance diversity and equality. However, in this thesis, we interpret temperature variation as a mechanism to explore the model's epistemic uncertainty.

2.2 Uncertainty in Machine Learning

As LLMs are increasingly deployed in real-world and high-stakes applications, for example in healthcare [Bed+25], their reliability becomes a critical concern. A fundamental challenge with deep neural networks, including LLMs, is that they are prone to overconfidence. They frequently assign high probabilities to incorrect predictions [Guo+17]. Consequently, evaluating a model's performance requires not only measuring its accuracy but also quantifying its **uncertainty**.

2.2.1 Aleatoric and Epistemic Uncertainty

In the field of machine learning, uncertainty can be categorized into two distinct sources: aleatoric and epistemic [KG17; HW21].

Aleatoric Uncertainty arises from the inherent complexity, noise, or randomness in the underlying data generation process. It is irreducable, meaning providing the model with more training data will not eliminate this uncertainty [KG17; HW21]. In the context of Natural Language Processing (NLP), aleatoric uncertainty is highly prevalant due to the inherent flexibility and ambiguity of human language. For a given prompt, there are often multiple semantically equivalent and perfectly valid responses. For example, the answers "It is raining heavily" and "It is pouring outside"

describe the same state, distributing the model's probability mass across different valid token sequences.

Epistemic uncertainty, on the other hand, stems from a lack of knowledge or information about the best model. This uncertainty occurs when a model encounters out-of-distribution (OOD) inputs or queries about facts absent from its training data. Unlike aleatoric uncertainty, epistemic uncertainty is reducable. Theoretically it can be minimized by providing the model with more comprehensive training data [KG17; HW21].

2.2.2 Standard Metrics: Shannon Entropy and Likelihood

To quantify the theoretical concepts of aleatoric and epistemic uncertainty, we rely on established mathematical metrics. In the context of probabilistic sequence modeling, two fundamental measures for evaluating a model's predictive confidence are **Likelihood** and **Shannon Entropy**.

Likelihood and Negative Log-Likelihood (NLL)
The likelihood represents the probability assigned by the model to a specific, observed sequence of tokens. For a given input context x and a target sequence $\mathbf{y}$, the sequence likelihood is the product of the individual token probabilities as defined in 2.1. Because multiplying many small probabilities leads to numerical underflow, it is standard practice to compute the **Negative Log-Likelihood (NLL)** [GBC16].

Taking the negative logarithm transforms the product of the probabilities into a sum. Furthermore, to fairly compare sequences of different lengths, the NLL can be normalized by the total number of tokens T, yielding the average NLL per token [MC18]:

$$\text{NLL}(\mathbf{y} \mid x) = -\sum_{t=1}^{T} \log P(y_t \mid \mathbf{y}_{<t}, x) \tag{2.8}$$

$$\text{average NLL}(\mathbf{y} \mid x) = -\frac{1}{T}\sum_{t=1}^{T} \log P(y_t \mid \mathbf{y}_{<t}, x) \tag{2.9}$$

A lower NLL indicates that the model asssigns a higher probability to the target sequence, implying higher confidence.

Shannon Entropy
While seqeuence likelihood evaluates the probability the model assigns to a *specific* target sequence, **Shannon Entropy** quantifies the fundamental uncertainty inherent in a probability distribution. In the context of autoregressive models, calculating the entropy of the predictive distribution over the entire vocabulary at a given time

step t measures the expected "surprise" or uncertainty in predicting the next token [Sha48].

For a discrete random variable Y_t representing the next token, with a probability distribution $P(Y_t \mid \mathbf{y}_{<t}, x)$ over the vocabulary $\mathcal{V}$, the Shannon Entropy H is defined as:

$$H(Y_t) = -\sum_{v \in \mathcal{V}} P(y_t = v \mid \mathbf{y}_{<t}, x) \log P(y_t = v \mid \mathbf{y}_{<t}, x) \tag{2.10}$$

Entropy provides a scalar summary of the distribution's shape.

- **Low Entropy** ($H \to 0$): The distribution is highly concentrated (peaked) on one or a few tokens, indicating that the model is highly certain about its next prediction.

- **High Entropy** ($H \to \log |\mathcal{V}|$): The distribution is flat and uniform across the vocabulary, indicating maximum uncertainty.

Limitations of Standard Metrics
Both NLL and Shannon Entropy are highly effective metrics, provided that the underlying probabiliy distribtuion P is perfectly calibrated and accurately reflects the true state of the world. However, these metrics inherently assume that the model's output distribution is a single, precise, and entirely correct representation of reality [Guo+17; GR07].

As demonstrated by Guo et al. [Guo+17], this assumption of a "precise probability" often breaks down in modern deep leraning models, including LLMs, which frequently suffer from severe overconfidence and calibration. Furthermore, standard precise probabilities mathematically struggle to distinguish between inherent data noise (aleatoric) and a fundamental lack of knowledge (epistemic) [HW21]. This failure to capture true epistemic misconfidence necessitates ore robust mathematical frameworks, such as Imprecise Probabilities and Credal Sets[Wal91].

2.3 Imprecise Probabilities and Credal Sets

Traditionally probability theory operates on the assumption that all forms of uncertainty can be adequately represented by a single, precise probability distribution [Wal91]. Under this paradigm, an agent or model is required to assign an exact numerical probability to every possible event, regardless of the amount of evidence available. While computationally convenient, this precise framework struggles to model severe epistemic uncertainty. When a model lacks sufficient training data or encounters an ambiguous query, forcing it to produce a single probability distribution often results in arbitrary or misleadingly confident predictions. Precise probabilities mathematically conflate the lack of evidence (epistemic uncertainty) with known

stochastic noise (aleatoric uncertainty), masking the model's true state of ignorance [HW21].

2.3.1 Introduction to Credal Sets

To address the limitations of precise probabilities, the framework of **imprecise probabilities** replaces the single distribution with a set of plausible distributions [Lev80; Wal91]. A closed, convex set of probability distributions is known as a **credal set** [Coz00], denoted by $\mathcal{K}$.

2.3.2 The Geometry of Credal Sets

A credal set $\mathcal{K}$ is geometrically represented as a closed, convex subset of the standard probability simplex Δ. For a classification task with K possible classes $c_1, c_2, \ldots, c_K$ (or semantic clusters), the standard simplex Δ_K is defined as [Pin88]:

$$\Delta_K = \left\{ \mathcal{P} \in \mathbb{R}^K \mid \forall k : \mathcal{P}_k \geq 0, \sum_{k=1}^{K} \mathcal{P}_k = 1 \right\} \tag{2.11}$$

In standard Bayesian inference, a precise probability distribution is a single point within this simplex. In contrast, a credal set represents a region of points.

Interval-Valued Probabilities
One of the most common and practical ways to define this region is through **probability intervals**. Instead of assigning a precise value $P(c_k)$, we assign a lower bound $\underline{P}(c_k)$ and an upper bound $\overline{P}(c_k)$ to each class c_k [CHM11].

The credal set induced by these intervals is the set of all distributions consistent with these bounds:

$$\mathcal{K} = \left\{ P \in \Delta_K \mid \underline{P}(c_k) \leq P(c_k) \leq \overline{P}(c_k), \sum_{k=1}^{K} P(c_k) = 1 \right\} \tag{2.12}$$

Geometrically, these linear constraints form a **convex polytope** within the simplex [CHM11].

If the intervals are wide (high epistemic uncertainty), the polytope covers a large volume of the simplex. If the intervals are narrow (low uncertainty), the polytopes

shrinks toward a single point. This geometric interpretation is fundamental to understanding how we quantify uncertainty in chapter 4.

2.3.3 Computing Uncertainty on Credal Sets

To quantify the total uncertainty of a model's prediction using a credal set, we must extend standard metrics like Shannon Entropy to operate over sets of distributions. Since we have a set of probability distributions $\mathcal{K}$, we can determine the **lower entropy** and **upper entropy** [AM05] by finding the minimum and maximum entropy values across the set [AKM06; HW21]:

$$H_*(\mathcal{K}) = \min_{q \in \mathcal{K}} H(q), \quad H^*(\mathcal{K}) = \max_{q \in \mathcal{K}} H(q) \tag{2.13}$$

We can use these bounds to quantify total uncertainty $U(\mathcal{K})$, aleatoric uncertainty $AU(\mathcal{K})$ and epistemic uncertainty $EU(\mathcal{K})$ [AKM06; HW21]:

$$U(\mathcal{K}) = AU(\mathcal{K}) + EU(\mathcal{K}) \tag{2.14}$$

$$H^*(\mathcal{K}) = H_*(\mathcal{K}) + (H^*(\mathcal{K}) - H_*(\mathcal{K})) \tag{2.15}$$

Here, upper entropy represents total uncertainty [AM05], lower entropy represents aleatoric uncertainty and the difference between them is used as a measure for epistemic uncertainty.

2.4 Relative Likelihood and Valid Temperature Sets

To systematically construct a credal set, we need a rigorous criterion to distinguish between plausubile and implausible probability distributions. Following recent advances in credal prediction [Löh+25], we achieve this by employing the statistical concept of **relative likelihood**.

2.4.1 Relative Likelihood

In standard maximum likelihood estimation, the goal is to find the single best parameter that maximizes the likelihood of the observed data. In the context of

our method, we treat the decoding temperature $\tau \in \mathcal{T}$ as the parameter of interest. Let $L(\tau)$ denote the likelihood of the target sequence generated at temperature τ. Furthermore, let τ^{ML} be the optimal temperature that yields the highest likelihood (or equivalently, minimizes the average Negative Log-Likelihood). To characterize the plausibility of any other temperature τ, we compute its relative likelihood $\gamma(\tau)$, defined as the ratio of its likelihood to the maximum likelihood [Bir62; Was90; WM99]:

$$\gamma(\tau) = \frac{L(\tau)}{L(\tau^{ML})} = \frac{L(\tau)}{\sup_{\tau' \in \mathcal{T}} L(\tau')} \tag{2.16}$$

By definition, $\gamma(\tau)$ is bounded in the interval $[0, 1]$, where $\gamma(\tau^{ML}) = 1$. The relative likelihood provides an intuitive measure of how well a specific temperature explains the data copmared to the optimal temperature setting.

2.4.2 Constructing α-Cuts for Temperatures

On the basis of the relative likelihood, a set of valid temperatures can be constructed by including only those that are plausible in the sense of surpassing a specific threshold $\alpha \in [0, 1]$. This set is formally referred to as an α**-cut** [ACC12]:

$$\mathcal{T}_\alpha = \{\tau \in \mathcal{T} : \gamma(\tau) \geq \alpha\} \tag{2.17}$$

According to this definiton, a temperature τ is considered implausible, and is therefore excluded from the valid set, if its likelihood is too small compared to the likelihood of the best temperature, specifically if it is less than α times the optimal likelihood.

Related Work

3

The reliable quantification of uncertainty in LLMs has garnered significant attention in recent years. As researchers seek to mitigate hallucinations and improve model safety, various metrics have been proposed to estimate when a model's generation is likely to be incorrect. This chapter reviews the evolution of these methods, starting from traditional uncertainty quantification (UQ) metrics, and culminating in state-of-the-art approaches like Semantic Entropy, which serves as the primary baseline for the credal prediction methodology developed in this thesis.

3.1 Traditional Uncertainty Quantification (UQ)

Applying traditional UQ techniques to LLMs presents substantial challenges. The computational overhead of Deep Ensembles [LPB17] or Bayesian frameworks like Monte Carlo dropout [GG16] makes them impractical for natural language generation (NLG) in massive models. Initial approaches to estimating LLM uncertainty directly adapted standard classificatoin metrics or relied on heuristic comparisons. These traditional baselines generally fall into three categories: predictive entropy, lexical similarity and self-evaluation.

Predictive Entropy Methods
The most straightforward methods evaluate the model's confidence using the probability distributions generated during decoding. Token-level entropy evaluates local uncertainty but fails to capture the holistic confidence of a factual claim. To address this, Malinin and Gales [MG21] proposed sequence-level predictive entropy (often implemented as length-normalized entropy), which estimates uncertainty across all possible output sequences. However, because these methods operate strictly on exact token matches, they artificially inflate uncertainty when an LLM distributes probability mass across multiple valid phrasings of the same answer.

Lexical Similarity
To account for the fact that generated answers might differ slightly in their suface forms, researchsers adapted translation metrics to quantify uncertainty. By sampling multiple answers for a given question, one can compute the average lexical similarity between them using metrics like ROUGE-L or BLEU [Fom+20]. If the sampled answers have high n-gram overlap, the model is deemed certain. While an improvement over strict sequence likelihood, lexical similarity still struggles with open-ended generation. An LLM might generate "Berlin" and "The capital of

Germany", which share zero lexical overlap (low ROUGE-L) but convey the exact same meaning, leading to false uncertainty signals.

Self-Evaluation
Another distinct approach treats the LLM as an independent evaluator of its own uncertainty. Methods like $p(\text{True})$ [Kad+22] prompt the model to evaluate its own generated answer or ask it directly if a proposed fact is correct. While intuitive, self-evaluation baselines are highly sensitive to the specific phrasing of the prompt.

All three of these traditional baseline families fail to reliably distinguish between benign linguistic flexibility and genuine factual ignorance. Traditional metrics treat these lexical variations as disjoint, competing hypotheses. This critical limitation necessitated a paradigm shift from lexical (word-based) probability to semantic (meaning-based) probability, leading to the development of Semantic Entropy [KGF23].

3.2 Semantic Entropy

To resolve the conflation of lexical flexibility and factual ignorance, Kuhn et al. [KGF23] introduced the concept of **Semantic Entropy (SE)**. Semantic Entropy shifts the unit of uncertainty measurement from the exact sequence of words to the underlying meaning, or *semantics*, of the generated text. Because SE serves as both the primary baseline and a core mechanical component of the methodology proposed in this thesis, we detail its formal computation below. The SE itself is computed in three steps:

1. Generating M sequences by the same model
2. Clustering the generated sequences by semantic equivalence
3. Computation of the semantic entropy

3.2.1 Semantic Clustering

The core premise of SE is that the infinite space of possible lexical sequences generated by an LLM can be partitioned into a finite set of discrete meanings.

Let x be the input prompt, and let $\mathcal{S} = \{s^{(1)}, s^{(2)}, \ldots, s^{(M)}\}$ be a set of M sequences generated by the model. We define a bidirectional semantic equivalence relation between sequences, denoted as $E(\cdot, \cdot)$, which holds true if sequence $s^{(i)}$ and sequence $s^{(j)}$ convey the exact same factual information, i.e., they entail, in the context of the prompt x. This equivalence relation groups the generated sequences into distinct

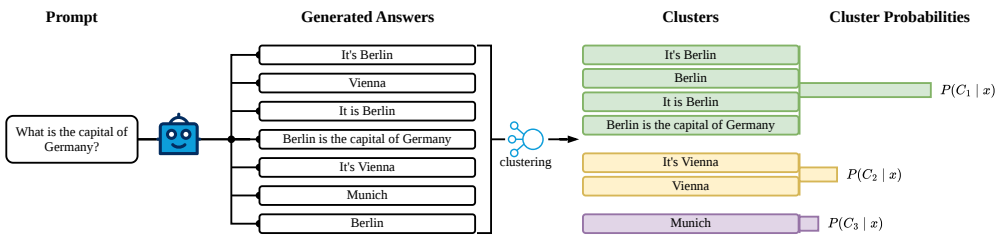

Fig. 3.1.: Visualization of Semantic Clustering for the question "What is the capital of Germany?". After the generation of seven answer strings, they are semantically clustered and afterwards the the cluster probabilities are computed as shown in 3.1.

semantic clusters (or equivalence classes), denoted as $C = \{c_1, c_2, \ldots, c_K\}$, where $\forall s^{(i)}, s^{(j)} \in c_k : E(s^{(i)}, s^{(j)})$ and $K \leq M$. In practice, this clustering is performed using an automated evaluation model, such as a Natural Language Inference (NLI) classifier (Kuhn et al. [KGF23] used the DeBERTa-large-MNLI model [He+21]).

3.2.2 Computing the Semantic Entropy

Once the sequences are clustered, the probability of a specific meaning (a cluster $c_k \in C$) is computed by marginalizing over the probabilities of all individual lexical sequences within that cluster. The cluster probability is given by:

$$P(c_k \mid x) = \sum_{s \in c_k} P(s \mid x) \tag{3.1}$$

By treating the clusters as the new categorical variables, the SE is calculated as the Shannon Entropy over the cluster probabilities:

$$\text{SE}(x) = -\sum_{k=1}^{K} P(c_k \mid x) \log P(c_k \mid x) \tag{3.2}$$

Because we can not know every possible meaning-class c, we can only sample c from the given distribution by the model. Thus Kuhn et al. [KGF23] use Monte Carlo integration over the semantic equivalence classes to estimate the expectation in equation 3.2:

$$\text{SE}(x) \approx -|C|^{-1} \sum_{i=1}^{|C|} \log P(C_i \mid x) \tag{3.3}$$

where

$$P(C_i \mid x) = \frac{P(c_i \mid x)}{\sum_{c \in C} P(c \mid x)} \tag{3.4}$$

SE represents a significant advancement over raw sequence likelihood. But it relies on a single, static probability distribution. In the standard implementation by Kuhn et al. [KGF23], sequences are sampled, and probabilities are evaluated at a single, fixed temperature (Kuhn et al. [KGF23] state that $\tau = 0.5$ is optimal).

As established in Section 2.3, precise probabilities are brittle when a model is miscalibrated or highly ignorant. This gap motivates the primary contribution of this thesis. Instead of computing SE on a single distribution, we propose computing it across a **Credal Set of distributions** generated by a valid range of temperatures. By analyzing how semantic cluster probabilities fluctate across this set, yielding upper and lower cluster bounds, we detect epistemic uncertainty that standard SE is blind to.

3.3 Credal Prediction based on Relative Likelihood

While SE addresses the issue of lexical equivalence in LLMs, it lacks a mechanism to handle severe epistemic uncertainty arising from a poorly calibrated, single probability distribution. To address this, we look to recent advancements in the field of imprecise probabilities, specifically the application of Credal Sets to deep learning.

3.3.1 Relative Likelihood Ensembles

Recently, Löhr et al. [Löh+25] proposed a theoretically grounded approach to credal prediction based on the statistical notion of relative likelihood. Instead of relying on a single maximum likelihood estimator (MLE), their framework defines the prediction target as a credal set induced by all plausible models whose relative likelihood exceeds a specific threshold α (the α-cut). Specifically, they introduce the Credal Relative Likelihood ($CreRL_\alpha$) method, which trains an ensemble of neural networks using an early-stopping strategy. Each model in the ensemble is trained until it reaches a sprecific relative likelihood threshold τ_i, ensuring that the final ensemble spans the entire valid α-cut. To guarantee diversity among the ensemble members, they further employ a novel "ToBias" initialization strategy, which initializes the models to predict degenerate distributions at the vertices of the probability simplex.

3.3.2 Bridging the Gap: From Ensembles to LLM Temperatures

The $CreRL_\alpha$ framework successfully demonstrates that relative likelihood can be used to construct robust credal sets, achieving superior coverage on standard classifiction benchmarks. However, this approach relies on optimizing model weights during training. Applying this weight-space ensemble approach directly to modern LLMs is computationally prohibitive. Training multiple copies of a multi-billion parameter model to convergence is infeasible for most applications. It also does not cleanly translate to the autoregressive, open-ended generation of sequences.

This limitation motivates the core innovation of our thesis. Shifting the the uncertainty quantification from the *weight space* (ensembling models) to the *decoding space* (ensmbling temperature).

4 Methodology: Credal Semantic Entropy via Temperature Scaling

This chapter introduces a novel methodology that constructs a credal set for LLMs by dynamically scaling the decoding temperature. By treating the temperature hyper-parameter as a continuous parameter space and appplying the statistical principles of relative likelihood, we systematically filter out implausible distributions. We then extend semantic entropy to operate over this credal set, yielding robust upper and lower uncertainty bounds that serve as a signal for hallucination detection.

4.1 Overview of the Pipeline

Our proposed methodology, which we term Credal Semantic Entopy, works for a given input prompt x (a question and optional context), in four sequential stages:

1. **Valid Temperature Selection (The α-cut)**: Before generation, we must determine which temperatures produce statistically valid probability distributions. Using a labeled dataset, we evaluate the model's average Negative Log-Likelihood (NLL) as shown in 2.9 across a discrete grid of temperatures. We identify the optimal temperature that minimizes the average NLL and compute the relative likelihood for all temperatures. The α-cut is constructed by all temperatures with a relative likelihood greater or equal than α.

2. **Generation and Semantic Clustering**: Using the prompt x, the LLM generates multiple candidate responses at every valid temperature in the α-cut. We cluster these generated sequences by their meaning over all valid temperatures.

3. **Computation Cluster Probability Bounds**: For each valid temperature, we compute the probability of each semantic cluster. Because the temperature modulates the entropy of the underlying token distributions, the probability of a given cluster will fluctate across the α-cut. By finding the minimum and maximum probability assigned to each cluster across all valid temperatures, we establish the lower and upper bounds for every semantic concept.

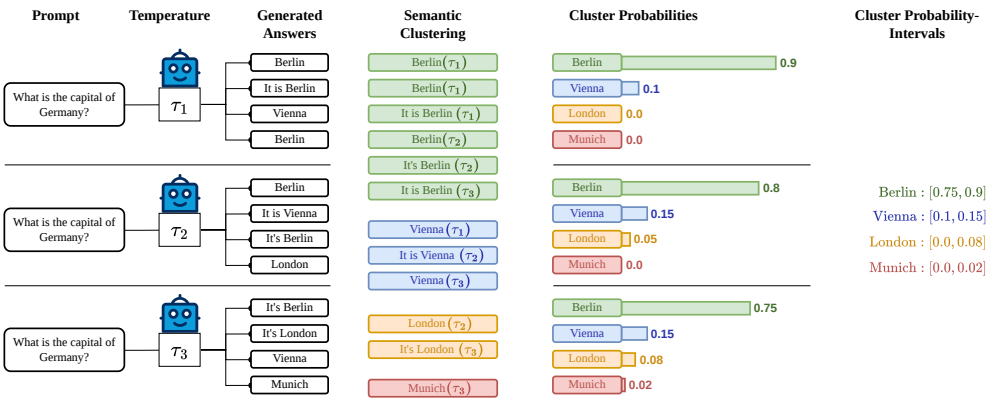

Fig. 4.1.: Visualization of the first three stages of our methodology. The process begins with multi-temperature generation, followed by semantic clustering performed on the union of all generated sequences. Finally, the probability mass of each cluster is marginalized at every temperature to establish the robust probability intervals used for uncertainty quantification.

4. **Quanifying Epistemic Uncertainty**: Finally, we compute the Upper Shannon Entropy using these bounds. This final scalar value is used to predict whether the model is hallucinating.

The following sections formally define the mathematical and computational mechanics of each stage in this pipeline.

4.2 Valid Temperature Selection (the $\alpha-$Cut)

We consider the task of free-form Question Answering (QA) using a pre-trained, autoregressive LLM $\mathcal{M}$ with a vocabulary $\mathcal{V}$. This process is performed a priori using a labeled training dataset or calibration dataset. We evaluate a predefined grid of candidate temperatures $\mathcal{T}$. For each temperature $\tau \in \mathcal{T}$, we measure its ability to assign high probability to the ground-truth answers. This involves three steps: computing the average NLL, deriving the relative likelihood and applying an α-cut threshold.

4.2.1 Computing the Average Negative Log-Likelihood

For every question-answer pair $(x^{(i)}, \mathbf{y}^{(i)})$ in the training dataset $\mathcal{D}_{\text{Train}}$ of size D, we compute the length-normalized NLL (2.9) of the ground-truth answer sequence under the temperature-scaled probability distribution P^{τ}. The average NLL for a given temperature τ across the entire dataset is computed as:

$$\overline{\text{NLL}}(\tau) = \frac{1}{D}\sum_{i=1}^{D}\left(-\frac{1}{T_i}\sum_{t=1}^{T_i}\log P^{\tau}(y_t^{(i)} \mid \mathbf{y}_{<t}^{(i)}, x^{(i)})\right) \tag{4.1}$$

where T_i is the length (number of tokens) of the i-th target sequence. This average NLL serves as our primary metric for evaluating how well a specific temperature calibrates the model's output distribution to the true data distribution. We denote the optimal temperature that yields the minimum NLL as τ^{ML}:

$$\tau^{ML} = \arg\min_{\tau \in \mathcal{T}} \overline{\text{NLL}}(\tau) \tag{4.2}$$

4.2.2 Deriving Relative Likelihoods

Because NLL values are absolute and can vary drastically depending on the model and dataset, it is difficult to define a universal threshold for what constitutes a "good" temperature. To standardize this, we map the average NLLs into the relative likelihood space.

As established in 2.16, the relative likelihood $\gamma(\tau)$ of any temperature is computed by comparing its likelihood to the optimal likelihood $L(\tau^{ML})$. In the logarithmic domain, this is equivalent to to the exponentiated difference of their average NLLs:

$$\gamma(\tau) = \exp\left(-\left(\overline{\text{NLL}}(\tau) - \overline{\text{NLL}}(\tau^{ML})\right)\right) \tag{4.3}$$

By definition, the optimal τ^{ML} has a relative likelihood of $\gamma(\tau^{ML}) = 1.0$. All other temperatures τ will have a relative likelihood $\gamma(\tau) \in [0, 1]$, explicitly quantifying how plausible they are compared to the best performing parameterization.

4.2.3 The α-Cut

Following the framework of credal prediction [Löh+25], we apply an α-cut, retaining only those temperatures whose relative likelihood meets or exceeds a specified confidence threshold $\alpha \in [0, 1]$. The final set of valid temperature is formally defined as:

$$\mathcal{T}_\alpha = \{\tau \in \mathcal{T} : \gamma(\tau) \geq \alpha\} \tag{4.4}$$

4.3 Generation and Semantic Clustering

Once the set of valid temperatures $\mathcal{T}_\alpha$ has been established via the train dataset, we apply the methodology to the unseen test queries. For a given input prompt x, the goal is map the model's raw token outputs into discrete semantic concepts. This requires a two-step process. Generating a diverse pool of candidate responses and clutsering them based on semantic equivalence.

4.3.1 Multi-Temperature Sequence Generation

The evaluate the model's uncertainty, we must sample responses from every valid temperature. For each temperature $\tau \in \mathcal{T}_\alpha$, we prompt the LLM $\mathcal{M}$ to generate N independent answer sequences.

Let $\mathcal{S}_\tau = \{s_1^\tau, s_2^\tau, \ldots, s_N^\tau\}$ denote the set of sequences generated at temperature τ. The total pool of generated responses for the prompt x is the union of all sequences generated across all valid temperatures:

$$\mathcal{S}_{total} = \bigcup_{\tau \in \mathcal{T}_\alpha} \mathcal{S}_\tau \tag{4.5}$$

By generating across $\mathcal{T}_\alpha$, we force the model to explore different regions of its probability space. Lower temperatures favor the most likely tokens, whereas higher temperatures flatten the distribution, encouraging the model to output lower-probability claims.

4.3.2 Bidirectional Entailment and Equivalence Classes

As discussed in Section 3.2, evaluating uncertainty on the raw lexical set $\mathcal{S}_{total}$ artificially inflates misconfidence due to the aleatoric linguistic variations. We therefore partition $\mathcal{S}_{total}$ into a set of semantic equivalence classes, or clusters, denoted as $C = \{c_1, c_2, \ldots, c_K\}$.

To determine if two distinct sequences s_i and s_j belong to the same cluster, we employ an automated Natural Language Inference (NLI) model (e.g., DeBERTa-large-MNLI [He+21]). The NLI model evaluates whether the premise sequence entails the hypothesis sequence. We define a strict semantic equivalence relation $E(\cdot, \cdot)$ which holds true if and only if there is **bidirectional entailment**: $(s_i \models s_j) \wedge (s_j \models s_i)$

If s_i entails s_j and s_j entails s_i, the sequences are deemed factually equivalent in the context of the prompt x and are assigned to the same cluster c_k. Thus, a cluster holds all sequences that belong to the same equivalence class of $E(\cdot, \cdot)$: $\forall s, s' \in c_k : E(s, s')$.

4.3.3 Constructing the Cluster Probability Distributions

Once the clusters C are established, we must compute the probability of each cluster under each temperature $\tau \in \mathcal{T}_\alpha$. The probability of a semantic cluster c_k at temperature τ is the normalized sum of the likelihoods of all sequences within that cluster, evaluated under the distribution P^τ:

$$P^\tau(C_k \mid x) = \frac{P^\tau(c_k \mid x)}{\sum_{c \in C} P^\tau(c \mid x)} \tag{4.6}$$

where

$$P^\tau(c_k \mid x) = \sum_{s \in c_k} P^\tau(s \mid x) \tag{4.7}$$

This marginalization step is crucial. It abstracts away the specific phrasing of the answers, yielding a set of discrete probability distributions $P^\tau(C)$ over the fundamental semantic meaning. It is important to note that the set of semantic clusters C is global. It contains every unique meaning generated across *all* valid temperatures. If a specific cluster c_k does not contain any sequences generated at a particular temperature τ, we assign it a zero probabilty mass for that distribution. This ensures that every distribution P^τ is defined over the same common support C, allowing for the direct element-wise comparison required to compute the upper and lower probability bounds in the next step.

4.3.4 Quantifying Credal Semantic Uncertainty

By scaling the temperature, we observe a range of probabilities for each semantic cluster. We utilize the observed probability fluctuations to establish upper and lower probability bounds for every semantic concept. The **Credal Set** is then formally defined as the set of *all* valid probability distributions that respect these element-wise bounds.

Establishing Probability Bounds
Let $C = \{c_1, c_2, \ldots, c_K\}$ be the set of identified semantic clusters. For each cluster

c_k, we inspect its probability mass across all valid temperatures $\tau \in \mathcal{T}_\alpha$ We define the **lower probability** $\underline{P}(c_k)$ and the **upper probability** $\overline{P}(c_k)$ as the minimum and maximum probabilities observed for that cluster.

$$\underline{P}(c_k) = \min_{\tau \in \mathcal{T}_\alpha} P^\tau(c_k \mid x) \qquad \overline{P}(c_k) = \max_{\tau \in \mathcal{T}_\alpha} P^\tau(c_k \mid x) \tag{4.8}$$

These bounds define an interval $[\underline{P}(c_k), \overline{P}(c_k)]$ representing the model's epistemic ambiguity regarding the plausibility of meaning c_k.

Defining the Credal Set
We construct the credal set $\mathcal{K}$ as the convex set of all probability distributions P over C that are consistent with these interval bounds. Formally:

$$\mathcal{K} = \left\{ P \in \Delta_K \mid \forall k : \underline{P}(c_k) \leq P(c_k) \leq \overline{P}(c_k), \sum_{k=1}^{K} P(c_k) = 1 \right\} \tag{4.9}$$

where Δ_K denotes the standard probability simplex. By defining $\mathcal{K}$ via these bounds, we capture not just the specific distributions observed at sampled temperatures, but the entire region of probability space supported by the valid relative likelihoods.

Upper Semantic Entropy (H^*)
This metric corresponds to the maximum entropy (highest uncertainty) attainable by any distribution within the credal set. It serves as our primary signal for hallucination detection 2.3.3, identifying cases where the model's interval constraints allow for a highly flat, uniform distribution:

$$H^*(x) = \max_{P \in \mathcal{K}} \left(-\sum_{k=1}^{K} P(c_k) \log P(c_k) \right) \tag{4.10}$$

Lower Semantic Entropy (H_*)
Conversely, the lower entropy corresponds to the most deterministic distribution consistent with the bounds:

$$H_*(x) = \min_{P \in \mathcal{K}} \left(-\sum_{k=1}^{K} P(c_k) \log P(c_k) \right) \tag{4.11}$$

The difference $H^* - H_*$ reflects the magnitude of epistemic uncertainty as shown in Section 2.3.3. For the experimental evaluation in this thesis, we utilize the upper semantic entropy H^*, as it provides the most conservative safety guarantee by considering the maximum plausible uncertainty.

5 Experiments

To validate the effectiveness of our proposed Credal Semantic Entropy method, we conducted a series of experiments on a standard free-form Question Answering (QA) benchmark. The primary objective of these experiments is to evaluate whether quantifying epistemic uncertainty via valid temperatures provides a more reliable signal for hallucination detection than standard single-point metrics.

5.1 Experimental Setup

5.1.1 Dataset: CoQA

We utilized the **Conversational Question Answering (CoQA)** dataset [RCM19] for our evaluation. CoQA is a large-scale dataset for building Conversational Question Answering systems. It contains 127k+ questions with answers obtained from 8k+ conversations about text passages from seven diverse domains [RCM19].

For our specific task of hallucination detection, we followed the experimental protocol established by Kuhn et al. [KGF23]. We used the development split of CoQA, which provides a diverse set of questions. The dataset was split into a **calibration set** (used to determine the valid temperature set) and a **test set** (used to evaluate the final uncertainty metrics).

5.1.2 Model Architecture

All experiments were conducted using the **Opt-6.7B** model [Zha+22]. OPT (Open Pre-trained Transfomer) is a decoder-only architecture that replicates the performance and behaviour of GPT-3 [Bro+20]. We chose the 6.7B parameter variant as it strikes a balance between computational feasibility and the capability to generate coherent text.

For the semantic clustering step, we employed a **DeBERTa-large-MNLI** model [He+21] to assess bidirectional entailment between generated answers.

5.1.3 Baselines

We compared our proposed **Credal Semantic Entropy (Credal SE)** against several strong baselines:

- **Length-Normalized NLL**: The average negative log-likelihood of the generated sequence, serving as the standard measure of model confidence.
- **Standard Semantic Entropy (SE)**: The original metric proposed by Kuhn et al. [KGF23],
- **Lexical Similarity (ROUGE-L)**: A metric that measures the lexical overlap between multiple sampled answers, serving as a proxy for certainty without semantic clustering [Lin04].
- **Predictive Entropy**: The standard entropy of the model's output distribution over the sequence of tokens. Unlike semantic entropy, this metric considers every lexical variation as a distinct outcome, calculating uncertainty directly on the raw strings without semantic clustering [MG21].

5.1.4 Evaluation Metric: AUROC

To quantify the performance of each uncertainty method, we treat hallucination detection as a binary classification problem. The core task is to determine if the uncertainty score can reliably distinguish between correct and incorrect model generations.

Ground Truth Labeling
To establish the ground-truth labels for our experiments, we compare the generated answers against the gold-standard reference answer provided in the CoQA dataset. We define correctness based on lexical overlap using the **ROUGE** metric (Recall-Oriented Understudy for Gisting Evaluation) [Lin04]. Specifically, a generated answer is labeled as:

- **Correct**: If the ROUGE-L score between the generation and the reference is strictly greater than 0.3.
- **Incorrect**: If the ROUGE-L score is less than or equal to 0.3.

This threshold allows for minor phrasing differences while ensuring that the core semantic content overlaps significantly with the reference.

AUROC Calculation
We report the **Area Under the Receiver Operating Characteristic cure (AUROC)** [Bra97]. The AUROC represents the probability that a randomly selected incorrect answer (hallucination) will have a higher uncertainty score than a randomly selected correct answer.

- AUROC $= 0.5$: Indicates random performance (no better than a coin flip)
- AUROC $= 1.0$: Indicates perfect detection (all hallucinations have higher uncertainty scores than all correct answers).

5.2 Results

Alpha	Beta	Gamma
0	1	2
3	4	5

Tab. 5.1.: This is a caption text.

6 Conclusion

6.1 System Section 1

6.2 System Section 2

6.3 Future Work

It would be interesting to see how this method performs on other tasks, like TriviaQA [Jos+17], SQuAD [RJL18] or some medical or legal dataset like medQA [Jin+20] where epistemic uncertainty carries higher risk and also when implemented using better models, like GPT 5 [Sin+25] or Gemini [Tea+25]. Another interesting strategy would be to not only focus on the temperature but on the decoding method itself. By sampling answers using Beam Search, Nucleus (Top-p) sampling [Hol+20], Top-k sampling [FLD18] and temperature sampling could also be another interesting method.

Appendix A

A.1 Experimental Details

Our implementation builds upon the codebase provided by Kuhn et al. [KGF23]. We utilize the HuggingFace Transformers library to implement both the generative model (OPT-2.7B, [Zha+22]) and the entailment model (DeBERTa-large-mnli, [He+21]).

To ensure a fair comparison, we replicate the generation parameters used in the baseline studies. The generation process is divided into two distinct phases using the `generate()` function:

Primary Answer Generation: To produce the model's "best guess" prediction (which is evaluated against the ground truth), we employ beam search with sampling enabled. Specifically, we set the beam size to 5 (`num_beams=5`) and enable multinomial sampling (`do_sample=True`).

Uncertainty Estimation Sampling: To generate the diverse set of sequences required for computing semantic entropy and our proposed credal metrics, we disable beam search (`num_beams=1`) and rely solely on multinomial sampling (`do_sample=True`). Following the findings of Kuhn et al. [KGF23], who demonstrated that uncertainty estimation performance saturates beyond a small number of samples, we generate $N = 10$ stochastic sequences per question.

The experiments run on one 80GB NVIDIA A100.

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Colophon

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